

SKILLWEEK5

TASK1: To Change Directories, Validate Paths, Update Working Directory, Handle Errors, Maintain Previous Directory, Test Navigation

```
GNU nano 7.2 task1.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <limits.h>

int main() {
    char current[PATH_MAX];
    char previous[PATH_MAX];

    getcwd(current, sizeof(current));
    strcpy(previous, current);

    printf("Current directory: %s\n", current);
    printf("Changing to /tmp...\n");

    if (chdir("/tmp") == 0) {
        strcpy(previous, current);

        getcwd(current, sizeof(current));

        printf("New directory: %s\n", current);
        printf("Previous directory: %s\n", previous);
    } else {
        perror("cd failed");
    }

    return 0;
}
```

Help **Write Out** **Where Is** **Cut** **Execute**
Exit **Read File** **Replace** **Paste** **Justify**

```
Hrushika Habeeb@Hrushika Habeeb: ~/
Hrushika Habeeb@Hrushika Habeeb:~$ mkdir Skills
Hrushika Habeeb@Hrushika Habeeb:~$ cd Skills
Hrushika Habeeb@Hrushika Habeeb:~/Skills$ nano task1.c
Hrushika Habeeb@Hrushika Habeeb:~/Skills$ gcc task1.c -o task1
```

```
Hrushika Habeeb@Hrushika Habeeb: ~/
Hrushika Habeeb@Hrushika Habeeb:~$ cd Skills
Hrushika Habeeb@Hrushika Habeeb:~/Skills$ nano task1.c
Hrushika Habeeb@Hrushika Habeeb:~/Skills$ gcc task1.c -o task1
Hrushika Habeeb@Hrushika Habeeb:~/Skills$ ./task1
Current directory: /home/hrushika_habeeb/Skills
Changing to /tmp...
New directory: /tmp
Previous directory: /home/hrushika_habeeb/Skills
Hrushika Habeeb@Hrushika Habeeb:~/Skills$ |
```

TASK2: To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic

```
jhansi_priya@Jhansi-Priya: ~/  × + ▾  
GNU nano 7.2  
#include <stdio.h>  
#include <stdlib.h>  
#include <string.h>  
#include <unistd.h>  
  
void pwd_command() {  
    char path[1024];  
  
    if (getcwd(path, sizeof(path)) != NULL)  
        printf("%s\n", path);  
    else  
        perror("pwd");  
}  
  
void cd_command(char *path) {  
    if (path == NULL) {  
        printf("cd: missing argument\n");  
        return;  
    }  
  
    if (chdir(path) != 0)  
        perror("cd");  
}  
  
void exit_command() {  
    printf("Exiting shell...\n");  
    exit(0);  
}  
  
int main() {  
    char input[100];  
    char *command;
```

```
jhansi_priya@Jhansi-Priya: ~/   
GNU nano 7.2  
char *command;  
char *argument;  
  
while (1) {  
    printf("myshell> ");  
    fgets(input, sizeof(input), stdin);  
  
    input[strcspn(input, "\n")] = '\0';  
  
    command = strtok(input, " ");  
    argument = strtok(NULL, " ");  
  
    if (command == NULL)  
        continue;  
  
    if (strcmp(command, "pwd") == 0) {  
        pwd_command();  
    }  
    else if (strcmp(command, "cd") == 0) {  
        cd_command(argument);  
    }  
    else if (strcmp(command, "exit") == 0) {  
        exit_command();  
    }  
    else {  
        printf("myshell: invalid command\n");  
    }  
}  
  
return 0;  
}
```

```
Hrushika Habeeb@Hrushika Habeeb:~/Skill5$ nano task2.c  
Hrushika Habeeb@Hrushika Habeeb:~/Skill5$ gcc task2.c  
Hrushika Habeeb@Hrushika Habeeb:~/Skill5$ gcc task2.c -o myshell  
Hrushika Habeeb@Hrushika Habeeb:~/Skill5$ ./myshell  
myshell> pwd  
/home/hrushika_habeeb/Skill5  
myshell> cd /tmp  
myshell> pwd  
/tmp  
myshell> hello  
myshell: invalid command  
myshell> exit  
Exiting shell...  
Hrushika Habeeb@Hrushika Habeeb:~/Skill5$ |
```

TASK3: To Retrieve Current Directory, Display Path, Process Exit Requests, Cleanup Resources, Save State, Verify Correct Behavior

```
jhansi_priya@Jhansi-Priya: ~/  ×  +  ∨
GNU nano 7.2 task3.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <limits.h>

void show_current_directory() {
    char path[PATH_MAX];

    if (getcwd(path, sizeof(path)) != NULL) {
        printf("Current Directory: %s\n", path);
    } else {
        perror("getcwd");
    }
}

int main() {

    printf("Testing pwd command:\n");
    show_current_directory();

    printf("\nTesting exit request...\n");

    printf("Resources cleaned successfully.\n");
    printf("Program exiting...\n");

    return 0;
}
```

```
Hrushika Habeeb@Hrushika Habeeb:~/Skills$ nano task3.c
Hrushika Habeeb@Hrushika Habeeb:~/Skill5$ gcc task3.c -o task3
Hrushika Habeeb@Hrushika Habeeb:~/Skill5$ ./task3
Testing pwd command:
Current Directory: /home/hrushika_habeeb/Skills

Testing exit request...
Resources cleaned succssfully.
Program exiting...
Hrushika Habeeb@Hrushika Habeeb:~/Skills$ |
```


TASK4: To Parse Export Syntax, Update Environment Variables, Validate Names, Handle Existing Variables, Export to Child Processes, Test Functionality

```
GNU nano 7.2 task4.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main() {

    char variable[] = "STUDENT";
    char value[] = "Jhansi";

    if (setenv(variable, value, 1) == 0) {
        printf("Variable exported successfully.\n");
    } else {
        perror("setenv");
        return 1;
    }

    printf("STUDENT = %s\n", getenv("STUDENT"));

    return 0;
}
```

```
GNU nano 7.2 child.c *
#include <stdio.h>
#include <stdlib.h>

int main() {

    char *value = getenv("STUDENT");

    if (value != NULL)
        printf("Child process received STUDENT = %s\n", value);
    else
        printf("STUDENT variable not found\n");

    return 0;
}
```

Program exiting....

```
hrushika_habeeb@Hrushika-Habeeb:~/Skill5$ nano task4.c
```

```
hrushika_habeeb@Hrushika-Habeeb:~/Skill5$ gcc task4.c -o task4
```

```
hrushika_habeeb@Hrushika-Habeeb:~/Skill5$ ./task4
```

Variable exported successfully.

STUDENT = Hrushika Habeeb

```
jhansi_priya@Jhansi-Priya: ~/ × + ▾
GNU nano 7.2 parent.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    setenv("STUDENT", "Jhansi", 1);
    printf("Parent: STUDENT = %s\n", getenv("STUDENT"));
    pid_t pid = fork();
    if (pid == 0) {
        execl("./child", "child", NULL);
        perror("execl");
        exit(1);
    }
    wait(NULL);
    return 0;
}
```

```
hrushika_habeeb@Hrushika-Habeeb:~/Skill5$ ./parent.c
-bash: ./child.c: Permission denied
hrushika_habeeb@Hrushika-Habeeb:~/Skill5$ nano parent.c
hrushika_habeeb@Hrushika-Habeeb:~/Skill5$ gcc parent.c -o parent
hrushika_habeeb@Hrushika-Habeeb:~/Skill5$ ./parent
Parent: STUDENT = Hrushika Habeeb
Child process received STUDENT = Hrushika Habeeb
hrushika_habeeb@Hrushika-Habeeb:~/Skill5$ |
```

